

# Probability Theory

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THIS LECTURE PROVIDES A FOUNDATIONAL OVERVIEW of key concepts in probability theory, emphasizing their relevance to statistical modeling and inference. It begins with the definition and properties of the expected value and variance, which quantify the central tendency and dispersion of random variables. The discussion then extends to joint probability distributions and joint cumulative distribution functions, capturing the behavior of multiple variables simultaneously. Concepts of conditional distribution and Bayes' theorem are introduced to formalize probabilistic reasoning under uncertainty and update beliefs based on observed evidence. The lecture concludes with an exploration of independence and conditional independence of random variables, highlighting their implications for model simplification and factorization in probabilistic frameworks. Together, these principles form the backbone of modern statistical analysis and machine learning, enabling rigorous interpretation and decision-making in complex systems.

## AGENDA:

- 1 Random vectors.
- 2 Joint probability distribution.
- 3 Marginal distribution
- 4 Examples.
- 5 Stochastic process definition.

## Expected Value and Variance of a Discrete Random Variable

For a discrete random variable  $X$ , its expected value is defined as:

$$\mu_X = E[X] = \sum_{\omega \in \Omega} X(\omega) p_X(\omega) = \sum_x x p(x).$$

For example, for the random variable  $K$  of Table 1, the expected value is

$$\begin{aligned} \mu_K &= 2\left(\frac{1}{36}\right) + 3\left(\frac{1}{18}\right) + 4\left(\frac{1}{12}\right) + 5\left(\frac{1}{9}\right) + 6\left(\frac{5}{36}\right) + 7\left(\frac{1}{6}\right) + 8\left(\frac{5}{36}\right) + 9\left(\frac{1}{9}\right) + 10\left(\frac{1}{12}\right) + 11\left(\frac{1}{18}\right) + 12\left(\frac{1}{36}\right) \\ &= 7. \end{aligned}$$

More generally, for a given function  $g(\cdot)$  we define:

$$\mu_{g(X)} = E[g(X)] = \sum_{\omega \in \Omega} g(X(\omega)) p_X(\omega) = \sum_x g(x) p(x).$$

The variance of  $X$  is defined as:

$$\text{var}[X] = \sum_{\omega \in \Omega} (X(\omega) - E[X])^2 p_X(\omega) = \sum_x (x - \mu_X)^2 p(x) = E[X^2] - (E[X])^2$$

For example, for the random variable  $K$  of Table 1, the variance is computed as follows

$$\begin{aligned} E[K^2] &= 2^2\left(\frac{1}{36}\right) + 3^2\left(\frac{1}{18}\right) + 4^2\left(\frac{1}{12}\right) + 5^2\left(\frac{1}{9}\right) + 6^2\left(\frac{5}{36}\right) + 7^2\left(\frac{1}{6}\right) + 8^2\left(\frac{5}{36}\right) + 9^2\left(\frac{1}{9}\right) + 10^2\left(\frac{1}{12}\right) + 11^2\left(\frac{1}{18}\right) + 12^2\left(\frac{1}{36}\right) \\ &= 54.83 \\ (E[K])^2 &= 49 \\ \text{var}[K] &= 5.83. \end{aligned}$$

### *Expected Value and Variance of a Continuous Random Variable*

For a continuous random variable  $X$  with probability density  $f_X(x)$ , its expected value is defined as:

$$\mu_X = E[X] = \int_{-\infty}^{\infty} \omega f_X(\omega) d\omega.$$

More generally, for a given function  $g(\cdot)$  we define:

$$E[g(X)] = \int_{-\infty}^{\infty} g(\omega) f_X(\omega) d\omega.$$

The variance of  $X$  is defined as:

$$\text{var}[X] = \int_{-\infty}^{\infty} (\omega - E[X])^2 f_X(\omega) d(\omega) = E[X^2] - (E[X])^2.$$

### *Joint Probability Distribution*

When dealing with multiple random variables, it is sometimes useful to use vector and matrix notations. Let  $X_1, X_2, \dots, X_N$  be  $N$  discrete random variables.

- The *joint probability function* of  $X_1, X_2, \dots, X_N$  is denoted as

$$p(x_1, x_2, \dots, x_N) := p_{X_1, X_2, \dots, X_N}(x_1, x_2, \dots, x_N) = P(X_1 = x_1, X_2 = x_2, \dots, X_N = x_N).$$

- Using vector notation we can write this distribution as

$$p(\mathbf{x}) := p_X(\mathbf{x}) = P(\mathbf{X} = \mathbf{x})$$

where  $\mathbf{X} = [X_1, X_2, \dots, X_N]^T$  is a column (random) vector having  $X_1, X_2, \dots, X_N$  as its components. Similarly  $\mathbf{x} = [x_1, x_2, \dots, x_N]^T$ . Note the abuse of notation in defining  $p(\mathbf{x})$  and  $p(x_1, x_2, \dots, x_N)$  above.

- Let  $X_1, X_2, \dots, X_N$  be a set of discrete random variables with joint distribution  $p(x_1, x_2, \dots, x_N)$ . Variable  $X_n$  can be *marginalized* from the joint distribution function as follows.

$$p(\mathbf{x}_{\neg n}) = p(x_1, x_2, \dots, x_{n-1}, x_{n+1}, \dots, x_N) = \sum_{x_n} p(x_1, x_2, \dots, x_N).$$

| $x_1$ | $x_2$ | $x_3$ | $p(x_1, x_2, x_3)$ |  | $x_1$ | $x_2$ | $p(x_1, x_2) = \sum_{x_3} p(x_1, x_2, x_3)$ |
|-------|-------|-------|--------------------|--|-------|-------|---------------------------------------------|
| 0     | 0     | 0     | $\theta_1$         |  | 0     | 0     | $\theta_1 + \theta_2$                       |
| 0     | 0     | 1     | $\theta_2$         |  | 0     | 1     | $\theta_3 + \theta_4$                       |
| 0     | 1     | 0     | $\theta_3$         |  | 1     | 0     | $\theta_5 + \theta_6$                       |
| 0     | 1     | 1     | $\theta_4$         |  | 1     | 1     | $\theta_7 + \theta_8$                       |
| 1     | 0     | 0     | $\theta_5$         |  |       |       |                                             |
| 1     | 0     | 1     | $\theta_6$         |  |       |       |                                             |
| 1     | 1     | 0     | $\theta_7$         |  |       |       |                                             |
| 1     | 1     | 1     | $\theta_8$         |  |       |       |                                             |

- Let  $X_1, X_2$  be two discrete random variables. The *conditional distribution function* of  $X_1$  given  $X_2$  is defined as

$$p(x_1 | x_2) = \frac{p(x_1, x_2)}{p(x_2)},$$

$$p(x_1, x_2) = p(x_1 | x_2)p(x_2) = p(x_2 | x_1)p(x_1),$$

with  $p(x_2) > 0$ .

| $x_1$ | $x_2$ | $p(x_1, x_2)$         |  | $x_2$ | $p(x_2) = \sum_{x_1} p(x_1, x_2)$           |
|-------|-------|-----------------------|--|-------|---------------------------------------------|
| 0     | 0     | $\theta_1 + \theta_2$ |  | 0     | $\theta_1 + \theta_2 + \theta_5 + \theta_6$ |
| 0     | 1     | $\theta_3 + \theta_4$ |  | 1     | $\theta_3 + \theta_4 + \theta_7 + \theta_8$ |
| 1     | 0     | $\theta_5 + \theta_6$ |  |       |                                             |
| 1     | 1     | $\theta_7 + \theta_8$ |  |       |                                             |

| $x_1$ | $x_2$ | $p(x_1   x_2)$                                                        |                               |
|-------|-------|-----------------------------------------------------------------------|-------------------------------|
| 0     | 0     | $(\theta_1 + \theta_2) / (\theta_1 + \theta_2 + \theta_5 + \theta_6)$ | $\sum_{x_1} p(x_1   x_2) = 1$ |
| 0     | 1     | $(\theta_3 + \theta_4) / (\theta_3 + \theta_4 + \theta_7 + \theta_8)$ |                               |
| 1     | 0     | $(\theta_5 + \theta_6) / (\theta_1 + \theta_2 + \theta_5 + \theta_6)$ |                               |
| 1     | 1     | $(\theta_7 + \theta_8) / (\theta_3 + \theta_4 + \theta_7 + \theta_8)$ |                               |

- Let  $X_1, X_2$ , and  $X_3$  be three discrete random variables. Compute  $p(x_2 | x_3)$  from the table given below. Compute  $\sum_{x_2} p(x_2 | x_3, x_1)$ .

| $x_2$ | $x_1$ | $x_3$ | $p(x_2   x_3, x_1)$ |  | $x_2$ | $x_3$ | $p(x_2   x_3)$              |
|-------|-------|-------|---------------------|--|-------|-------|-----------------------------|
| 0     | 0     | 0     | $\theta_1$          |  | 0     | 0     | $\theta_1 + \theta_3$       |
| 0     | 0     | 1     | $\theta_2$          |  | 0     | 1     | $\theta_2 + \theta_4$       |
| 0     | 1     | 0     | $\theta_3$          |  | 1     | 0     | $1 - (\theta_1 + \theta_3)$ |
| 0     | 1     | 1     | $\theta_4$          |  | 1     | 1     | $1 - (\theta_2 + \theta_4)$ |
| 1     | 0     | 0     | $1 - \theta_1$      |  |       |       |                             |
| 1     | 0     | 1     | $1 - \theta_2$      |  |       |       |                             |
| 1     | 1     | 0     | $1 - \theta_3$      |  |       |       |                             |
| 1     | 1     | 1     | $1 - \theta_4$      |  |       |       |                             |

- Let  $\mathbf{X} = \{X_1, X_2, \dots, X_N\}$ , and  $\mathbf{Y} = \{Y_1, Y_2, \dots, Y_M\}$  be two sets of discrete random variables. The *conditional distribution function* of  $\mathbf{X}$

given  $\mathbf{Y}$  is defined as

$$p(\mathbf{x} | \mathbf{y}) = \frac{p(\mathbf{x}, \mathbf{y})}{p(\mathbf{y})}$$

with  $p(\mathbf{y}) > 0$ .

- Let  $X_1, X_2$  be two discrete random variables. The Bayes rule establishes that

$$p(x_1 | x_2) = \frac{p(x_1, x_2)}{p(x_2)} = \frac{p(x_2 | x_1)p(x_1)}{p(x_2)} = \frac{p(x_2 | x_1)p(x_1)}{\sum_{x_1} p(x_2 | x_1)p(x_1)}$$

with  $p(x_2) > 0$ .

- Let  $\mathbf{X} = \{X_1, X_2, \dots, X_N\}$ , and  $\mathbf{Y} = \{Y_1, Y_2, \dots, Y_M\}$  be two sets of discrete random variables. The *Bayes rule* establishes that

$$p(\mathbf{x} | \mathbf{y}) = \frac{p(\mathbf{x}, \mathbf{y})}{p(\mathbf{x})} = \frac{p(\mathbf{y} | \mathbf{x})p(\mathbf{x})}{p(\mathbf{y})} = \frac{p(\mathbf{y} | \mathbf{x})p(\mathbf{x})}{\sum_{\mathbf{x}} p(\mathbf{y} | \mathbf{x})p(\mathbf{x})}$$

with  $p(\mathbf{y}) > 0$ .

- Let  $X_1, X_2$  be two discrete random variables. These variables are independent if

$$p(x_1, x_2) = p(x_1)p(x_2),$$

or alternatively,

$$p(x_1 | x_2) = p(x_1).$$

- Let  $\mathbf{X} = \{X_1, X_2, \dots, X_N\}$ , and  $\mathbf{Y} = \{Y_1, Y_2, \dots, Y_M\}$  be two sets of discrete random variables. These sets of variables are *independent* if

$$p(\mathbf{x}, \mathbf{y}) = p(\mathbf{x})p(\mathbf{y}),$$

or alternatively,

$$p(\mathbf{x} | \mathbf{y}) = p(\mathbf{x}).$$

- Let  $X_1, X_2, X_3$  be three discrete random variables. We say that  $X_1$  is *conditionally independent* of  $X_2$  given  $X_3$  (denoted by  $X_1 \perp\!\!\!\perp X_2 | X_3$ ) if

$$p(x_1, x_2 | x_3) = p(x_1 | x_3)p(x_2 | x_3)$$

or alternatively,

$$p(x_1 | x_2, x_3) = p(x_1 | x_3).$$

- Let  $\mathbf{X} = \{X_1, X_2, \dots, X_N\}$ ,  $\mathbf{Y} = \{Y_1, Y_2, \dots, Y_M\}$  and  $\mathbf{Z} = \{Z_1, Z_2, \dots, Z_L\}$  be three sets of discrete random variables. We say that  $\mathbf{X}$  is *conditionally independent* of  $\mathbf{Y}$  given  $\mathbf{Z}$  (denoted by  $\mathbf{X} \perp\!\!\!\perp \mathbf{Y} | \mathbf{Z}$ ) if

$$p(\mathbf{x}, \mathbf{y} | \mathbf{z}) = p(\mathbf{x} | \mathbf{z})p(\mathbf{y} | \mathbf{z})$$

or alternatively,

$$p(\mathbf{x} | \mathbf{y}, \mathbf{z}) = p(\mathbf{x} | \mathbf{z}).$$

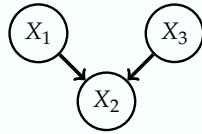
- *Bayesian networks in brief* A Bayesian network is a probabilistic graphical model that represents a set of variables and their conditional dependencies via a directed acyclic graph (DAG). Bayesian networks are directed acyclic graphs (DAGs) whose nodes represent variables in the Bayesian sense: they may be observable quantities, latent variables, unknown parameters, or hypotheses. Each edge represents a direct conditional dependency. Any pair of nodes that are not connected (i.e. no path connects one node to the other) represent variables that are conditionally independent of each other. Each node is associated with a probability function that takes, as input, a particular set of values for the node's parent variables and gives (as output) the probability (or probability distribution, if applicable) of the variable represented by the node. Bayesian networks having  $X_1, X_2, \dots, X_N$  variables factorize as

$$p(x_1, x_2, \dots, x_N) = \prod_{n=1}^N p(x_n | \text{pa}(X_n)),$$

where  $\text{pa}(x_n)$  represent the parents of variable  $X_n$  in the associated DAG.

#### Box 1: Bayesian Network Example

Consider the example of a Bayesian network having three random variables  $X_1, X_2, X_3$ .



$$p(x_1, x_2, x_3) = p(x_2 | x_1, x_3) p(x_3) p(x_1)$$

| $x_2$ | $x_1$ | $x_3$ | $p(x_2   x_3, x_1)$ | $x_1$ | $p(x_1)$       |
|-------|-------|-------|---------------------|-------|----------------|
| 0     | 0     | 0     | $\theta_1$          | 0     | $\theta_5$     |
| 0     | 0     | 1     | $\theta_2$          | 1     | $1 - \theta_5$ |
| 0     | 1     | 0     | $\theta_3$          | $x_3$ | $p(x_3)$       |
| 0     | 1     | 1     | $\theta_4$          | 0     | $\theta_6$     |
| 1     | 0     | 0     | $1 - \theta_1$      | 1     | $1 - \theta_6$ |
| 1     | 0     | 1     | $1 - \theta_2$      |       |                |
| 1     | 1     | 0     | $1 - \theta_3$      |       |                |
| 1     | 1     | 1     | $1 - \theta_4$      |       |                |

**Definition 1: A Bayesian network structure**

A Bayesian network structure  $G$  is a directed acyclic graph whose nodes represent random variables  $X_1, \dots, X_n$ . Let  $\text{pa}(X_i)$  denote the parents of  $X_i$  in  $G$ , and let  $\text{nd}(X_i)$  denote the variables in the graph that are not descendants of  $X_i$ . Then  $G$  encodes the following set of conditional independence assumptions, called the *local independencies*:

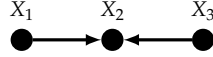
$$\forall X_i \in \{X_1, \dots, X_n\}, \quad X_i \perp\!\!\!\perp \text{nd}(X_i) \mid \text{pa}(X_i)$$

- Bayesian Network Examples**

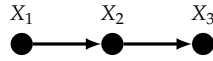
In other words, the local independencies state that each node  $X_i$  is conditionally independent of its nondescendants given its parents. Solution.

Draw the DAG associated with the following probability distributions:

1.  $p(x_1, x_2, x_3) = p(x_2)p(x_3)p(x_2 \mid x_1, x_3)$ . Show that  $X_1 \perp\!\!\!\perp X_3$ .

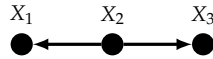


2.  $p(x_1, x_2, x_3) = p(x_1)p(x_2|x_1)p(x_3|x_2)$ . Show that  $X_1 \perp\!\!\!\perp X_3|X_2$ .



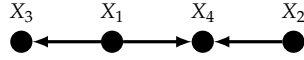
$$\begin{aligned}
 p(x_1, x_3 \mid x_2) &= \frac{p(x_1, x_2, x_3)}{p(x_2)} \\
 &= \frac{p(x_1)p(x_2|x_1)p(x_3|x_2)}{p(x_2)} \\
 &= \frac{p(x_1, x_2)p(x_3|x_2)}{p(x_2)} \\
 &= \frac{p(x_1 \mid x_2)p(x_2)p(x_3|x_2)}{p(x_2)} \\
 &= p(x_1|x_2)p(x_3|x_2) \\
 &\implies X_1 \perp\!\!\!\perp X_3|X_2
 \end{aligned}$$

3.  $p(x_1, x_2, x_3) = p(x_2)p(x_1|x_2)p(x_3|x_2)$ . Show that  $X_1 \perp\!\!\!\perp X_3|X_2$ .

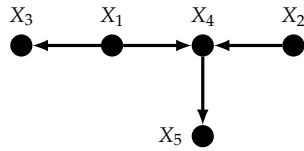


$$\begin{aligned}
 p(x_1, x_3 \mid x_2) &= \frac{p(x_1, x_2, x_3)}{p(x_2)} \\
 &= \frac{p(x_2)p(x_1|x_2)p(x_3|x_2)}{p(x_2)} \\
 &= p(x_1|x_2)p(x_3|x_2) \\
 &\implies X_1 \perp\!\!\!\perp X_3 \mid X_2
 \end{aligned}$$

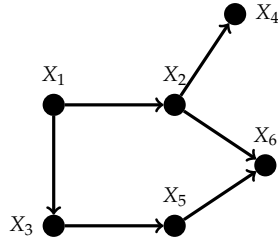
4.  $p(x_1, x_2, x_3, x_4) = p(x_1)p(x_2)p(x_3|x_1)p(x_4|x_1, x_2)$ . Show that  $X_3 \perp\!\!\!\perp X_4 \mid X_1$ .



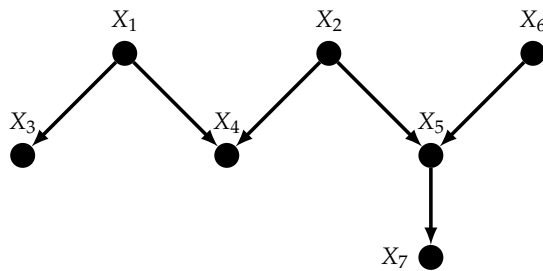
5.  $p(x_1, x_2, x_3, x_4, x_5) = p(x_1)p(x_2)p(x_3|x_1)p(x_4|x_1, x_2)p(x_5|x_4)$ . Show that  $X_1 \perp\!\!\!\perp X_5 \mid X_4$ .



6.  $p(x_1, x_2, x_3, x_4, x_5, x_6) = p(x_1)p(x_2|x_1)p(x_3|x_1)p(x_4|x_2)p(x_5|x_3)p(x_6|x_2, x_5)$ . Show that  $X_2 \perp\!\!\!\perp X_3 \mid X_1$ .



7. Write down the factorization of  $p(x_1, x_2, x_3, x_4, x_5, x_6, x_7)$  encoded in the following DAG.



### Box 2: Application Example (Was it the burglar?)

Mary lives in San Francisco City. One afternoon, she is driving back home and receives a phone call from her neighbor Jane. She told her that her house alarm was set off (A). While driving, she also heard on the radio (R) that a small earthquake (E) hit the city. Small earthquakes sometimes activate the alarm, and perhaps this is the reason why the alarm was sounding.

- What is the probability that a burglar (B) broke into the house?
- What is the probability that a burglar broke into the house given that the alarm was set off?
- What is the probability that a burglar broke into the house given that the alarm was set off and a small earthquake hit the city?

To solve this problem we make several considerations.

1. Define the random variables  $R(R) = 1, R(R^c) = 0, A(A) = 1, A(A^c) = 0, E(E) = 1, E(E^c) = 0$ , and  $B(B) = 1, B(B^c) = 0$ .
2. Assume that the joint distribution of random variables  $R, A, E$ , and  $B$  factorizes as

$$p(R, A, E, B) = p(E = e)p(B = b)p(A = a | B = b, E = e)p(R = r | E = e)$$

our using our simplified notation

$$p(r, a, e, b) = p(e)p(b)p(a | b, e)p(r | e).$$

The factors of the probability distribution correspond to the following tables.

| $b \quad p(b)$ |   |            |
|----------------|---|------------|
| $B^c$          | 0 | $b_1 = .9$ |
| B              | 1 | $1 - b_1$  |

| $e \quad p(e)$ |   |              |
|----------------|---|--------------|
| $E^c$          | 0 | $e_1 = 0.95$ |
| E              | 1 | $e_2$        |

| $r \quad p(r)$ |   |              |
|----------------|---|--------------|
| $R^c$          | 0 | $c_1 = 0.05$ |
| R              | 1 | $c_2$        |

| $b \quad e \quad p(b, e)$ |       |   |   |           |
|---------------------------|-------|---|---|-----------|
| $B^c$                     | $E^c$ | 0 | 0 | $b_1 e_1$ |
| $B^c$                     | E     | 0 | 1 | $b_1 e_2$ |
| B                         | $E^c$ | 1 | 0 | $b_2 e_1$ |
| B.                        | E     | 1 | 1 | $b_2 e_2$ |

| $r \quad e \quad p(r   e)$ |       |   |   |                 |
|----------------------------|-------|---|---|-----------------|
| $R^c$                      | $E^c$ | 0 | 0 | $f_1 = 0.99$    |
| $R^c$                      | E     | 0 | 1 | $f_2 = 0.01$    |
| R                          | $E^c$ | 1 | 0 | $f_3 = 1 - f_1$ |
| R.                         | E     | 1 | 1 | $f_4 = 1 - f_2$ |

|       |       |       | $a$ | $b$ | $e$ | $p(a   b, e)$ |                                 |
|-------|-------|-------|-----|-----|-----|---------------|---------------------------------|
| $A^c$ | $B^c$ | $E^c$ | 0   | 0   | 0   | $q_1$         | $q_1 + q_5 = 1, q_2 + q_6 = 1,$ |
| $A^c$ | $B^c$ | $E$   | 0   | 0   | 1   | $q_2$         | $q_3 + q_5 = 1, q_4 + q_8 = 1,$ |
| $A^c$ | $B$   | $E^c$ | 0   | 1   | 0   | $q_3$         | $b_1 + b_2 = 1, e_1 + e_2 = 1,$ |
| $A^c$ | $B$   | $E$   | 0   | 1   | 1   | $q_4$         | $c_1 + c_2 = 1,$                |
| $A$   | $B^c$ | $E^c$ | 1   | 0   | 0   | $q_5$         | $f_1 + f_3 = 1,$                |
| $A$   | $B^c$ | $E$   | 1   | 0   | 1   | $q_6$         | $f_2 + f_4 = 1.$                |
| $A$   | $B$   | $E^c$ | 1   | 1   | 0   | $q_7$         |                                 |
| $A$   | $B$   | $E$   | 1   | 1   | 1   | $q_8$         |                                 |

The table representing the joint probability function is

|       |       |       | $r$   | $a$ | $b$ | $e$ | $p(r, a, e, b)$ |                            |
|-------|-------|-------|-------|-----|-----|-----|-----------------|----------------------------|
| $R^c$ | $A^c$ | $B^c$ | $E^c$ | 0   | 0   | 0   | 0               | $p_1 = b_1 e_1 f_1 p_1$    |
| $R^c$ | $A^c$ | $B^c$ | $E$   | 0   | 0   | 0   | 1               | $p_2 = b_1 e_2 f_1 p_2$    |
| $R^c$ | $A^c$ | $B$   | $E^c$ | 0   | 0   | 1   | 0               | $p_3 = b_2 e_1 f_1 p_3$    |
| $R^c$ | $A^c$ | $B$   | $E$   | 0   | 0   | 1   | 1               | $p_4 = b_2 e_2 f_1 p_4$    |
| $R^c$ | $A$   | $B^c$ | $E^c$ | 0   | 1   | 0   | 0               | $p_5 = b_1 e_1 f_2 p_5$    |
| $R^c$ | $A$   | $B^c$ | $E$   | 0   | 1   | 0   | 1               | $p_6 = b_1 e_2 f_2 p_6$    |
| $R^c$ | $A$   | $B$   | $E^c$ | 0   | 1   | 1   | 0               | $p_7 = b_2 e_1 f_2 p_7$    |
| $R^c$ | $A$   | $B$   | $E$   | 0   | 1   | 1   | 1               | $p_8 = b_2 e_2 f_2 p_8$    |
| $R$   | $A^c$ | $B^c$ | $E^c$ | 1   | 0   | 0   | 0               | $p_9 = b_1 e_1 f_3 p_1$    |
| $R$   | $A^c$ | $B^c$ | $E$   | 1   | 0   | 0   | 1               | $p_{10} = b_1 e_2 f_3 p_2$ |
| $R$   | $A^c$ | $B$   | $E^c$ | 1   | 0   | 1   | 0               | $p_{11} = b_2 e_1 f_3 p_3$ |
| $R$   | $A^c$ | $B$   | $E$   | 1   | 0   | 1   | 1               | $p_{12} = b_2 e_2 f_3 p_4$ |
| $R$   | $A$   | $B^c$ | $E^c$ | 1   | 1   | 0   | 0               | $p_{13} = b_1 e_1 f_4 p_5$ |
| $R$   | $A$   | $B^c$ | $E$   | 1   | 1   | 0   | 1               | $p_{14} = b_1 e_2 f_4 p_6$ |
| $R$   | $A$   | $B$   | $E^c$ | 1   | 1   | 1   | 0               | $p_{15} = b_2 e_1 f_4 p_7$ |
| $R$   | $A$   | $B$   | $E$   | 1   | 1   | 1   | 1               | $p_{16} = b_2 e_2 f_4 p_8$ |

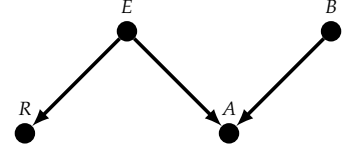


Figure 1: Directed Acyclic Graph encoding the probability distribution  $p(r, a, e, b) = p(e)p(b)p(a | b, e)p(r | e)$ . Dark circles represent random variables  $R, A, E$ , and  $B$ .

- The *graphical model* (a Directed Acyclic Graph or DAG) encoding the probability distribution factorization is shown in Figure 21.
- Assume that  $q_1 = 0.98, q_2 = 0.9, q_3 = 0.1, q_4 = 0.01$ .
- What is the probability that a burglar broke into the house?

$$P(B = 1) = P(B) = p_3 + p_4 + p_7 + p_8 + p_{11} + p_{12} + p_{15} + p_{16},$$

$$P(B = 1) = \sum_a \sum_r \sum_e p(r, a, B = 1, e) = 0.1.$$

- What is the probability that a burglar broke into the house given

that the alarm was set off?

$$P(B = 1 \mid A = 1) = \frac{P(B = 1, A = 1)}{P(A = 1)}$$

$$= \frac{p_7 + p_8 + p_{15} + p_{16}}{p_5 + p_6 + p_7 + p_8 + p_{13} + p_{14} + p_{15} + p_{16}},$$

$$P(B = 1 \mid A = 1) = \frac{\sum_r \sum_e p(r, A = 1, B = 1, e)}{\sum_r \sum_b \sum_e p(r, A = 1, b, e)} = 0.81.$$

- What is the probability that a burglar broke into the house given that the alarm was set off and a small earthquake hit the city?

$$P(B = 1 \mid A = 1, E = 1) = \frac{P(B = 1, A = 1, E = 1)}{P(A = 1, E = 1)}$$

$$= \frac{p_8 + p_{16}}{p_6 + p_8 + p_{14} + p_{16}},$$

$$P(B = 1 \mid A = 1, E = 1) = \frac{\sum_r p(r, A = 1, B = 1, E = 1)}{\sum_r \sum_b p(r, A = 1, b, E = 1)}$$

$$= 0.51.$$

### Box 3: Solution Summary

- What is the probability that a burglar broke into the house?

$$P(B = 1) = 0.1.$$

- What is the probability that a burglar broke into the house given that the alarm was set off?

$$P(B = 1 \mid A = 1) = 0.81.$$

- What is the probability that a burglar broke into the house given that the alarm was set off and a small earthquake hit the city?

$$P(B = 1 \mid A = 1, E = 1) = 0.51.$$

- The *a priori* probability that the burglar broke into the house is 0.1. The probability that a burglar broke into the house given that the alarm was set off is much higher (0.81). However, knowing that a small earthquake hit the city *explains away* the observation that the alarm was set off, diminishing the probability that a burglar broke (0.51).

- Note that the factorization features of the probability distribution and the sum-product distributive property help us reduce the

computational complexity.

$$\begin{aligned}
 p(b \mid a) &= \frac{p(a, b)}{p(a)} = \frac{\sum_r \sum_e p(r, a, e, b)}{\sum_b \sum_r \sum_e p(r, a, e, b)} \\
 &= \frac{\sum_r \sum_e p(e) p(b) p(a \mid e, b) p(r \mid e)}{\sum_b \sum_r \sum_e p(e) p(b) p(a \mid e, b) p(r \mid e)} \\
 &= \frac{p(b) \sum_e p(e) p(a \mid e, b) \sum_r p(r \mid e)}{\sum_b p(b) \sum_e p(e) p(a \mid e, b) \sum_r p(r \mid e)} \\
 &= \frac{p(b) \sum_e p(e) p(a \mid e, b) \phi_1(e)}{\sum_b p(b) \sum_e p(e) p(a \mid e, b) \phi_1(e)} \\
 &= \frac{p(b) \phi_2(a, b)}{\sum_b p(b) \phi_2(a, b)} \\
 &= \frac{p(b) \phi_2(a, b)}{\phi_3(a)}
 \end{aligned}$$

#### Box 4: Application Example (Pairs of dice)

You are told that there are two pairs of coins. The first pair is fair,

$$\theta_1(H) = \theta_1(T) = \frac{1}{2}.$$

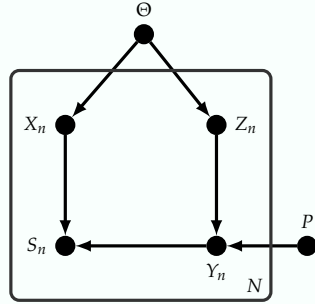
The second pair is biased as both coins have probability

$$\theta_2(H) = \frac{2}{3}, \theta_2(T) = \frac{1}{3}$$

of producing heads and tails, respectively. One of the two pairs is chosen at random with probability 0.5 and thrown 10 times. The sum of the coins is recorded for each throw.

1. If the throw results in the sequence "1010101010"? Which pair of dice was picked for the throw?
2. Repeat (1) for the sequence "2222200000".
3. Assume that the result of the second coin is flipped (F) with probability  $p = 0.5$  before recording the result of the throw with probability. Repeat 1 and 2 under this assumption. Consider the case for which  $p = 0.2$ . Repeat exercises 1 and 2.

The graphical model associated with this problem is as follows.



- $X_n$  represents the  $n$ -th result from the throw of the first coin;  $X_n(H) = 1$  and  $X_n(T) = 0$ , and

$$p(x_n | \theta) = \theta^{x_n} (1 - \theta)^{(1-x_n)}.$$

- $Z_n$  represents the unobserved result from the  $n$ -th throw of the second coin;  $Z_n(F) = 1$  and  $Z_n(F^c) = 0$ , and

$$p(z_n | \theta) = \theta^{z_n} (1 - \theta)^{(1-z_n)}.$$

- $Y_n$  represents the recorded result from the  $n$ -th throw of the second coin;  $Y_n(H) = 1$  and  $Y_n(T) = 0$ , and

$$p(y_n | z_n, p) = p^{I(y_n \neq z_n)} (1 - p)^{I(y_n = z_n)}.$$

- $S_n$  is a deterministic function of  $X_n$  and  $Y_n$ ,

$$S_n = X_n + Y_n,$$

$$p(s_n | x_n, y_n) = I(s_n == x_n + y_n),$$

and

$$I(e) = \begin{cases} 1, & \text{if } e = \text{true;} \\ 0, & \text{otherwise.} \end{cases}$$

Let  $\mathbf{X} = (X_1, X_2, \dots, X_N)$ ,  $\mathbf{Y} = (Y_1, Y_2, \dots, Y_N)$ ,  $\mathbf{Z} = (Z_1, Z_2, \dots, Z_N)$ ,  $\mathbf{S} = (S_1, S_2, \dots, S_N)$ ,  $f_n = I(y_n \neq z_n) = 1 - I(y_n = z_n)$ , and  $\mathbf{f} = (f_1, f_2, \dots, f_N)$ . ( $f_n$  indicates whether the second was flipped or not.) The joint probability distribution for  $\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s}$  given  $\theta$  and  $p$  is

$$\begin{aligned}
 p(\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s}, \theta, p) &= p(\theta)p(p) \prod_{n=1}^N p(x_n | \theta)p(z_n | \theta)p(y_n | z_n, p)p(s_n | x_n, y_n) \\
 p(\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s} | \theta, p) &= \frac{p(\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s}, \theta, p)}{p(\theta)p(p)} \\
 p(\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s} | \theta, p) &= \prod_{n=1}^N p(x_n | \theta)p(z_n | \theta)p(y_n | z_n, p)p(s_n | x_n, y_n) \\
 &= \prod_{n=1}^N \theta^{x_n} (1 - \theta)^{(1-x_n)} \theta^{z_n} (1 - \theta)^{(1-z_n)} p^{I(y_n \neq z_n)} (1 - p)^{I(y_n = z_n)} I(s_n = x_n + y_n). \\
 &= \theta^{\sum_{n=1}^N x_n} (1 - \theta)^{\sum_{n=1}^N (1-x_n)} \theta^{\sum_{n=1}^N z_n} (1 - \theta)^{\sum_{n=1}^N (1-z_n)} p^{\sum_{n=1}^N I(y_n \neq z_n)} (1 - p)^{\sum_{n=1}^N I(y_n = z_n)}
 \end{aligned}$$

Notice that

$$\begin{aligned}
 x_n &= s_n - y_n, \\
 y_n &= z_n(1 - f_n) + (1 - z_n)f_n,
 \end{aligned}$$

and

$$x_n = s_n - z_n + 2z_nf_n - f_n.$$

We define  $N_x$ , and  $N_z$  as follows,

$$N_x = \sum_{n=1}^N (s_n - z_n + 2z_nf_n - f_n) = N_s - N_z + 2N_{zf} - N_f \geq 0,$$

$$N_s = \sum_{n=1}^N s_n,$$

$$N_z = \sum_{n=1}^N z_n,$$

$$N_{zf} = \sum_{n=1}^N z_nf_n,$$

and

$$N_f = \sum_{n=1}^N f_n.$$

Therefore we can write the probability distribution function as follows,

$$\begin{aligned}
 p(\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s} \mid \theta, p) &= \theta^{\sum_{n=1}^N x_n} (1 - \theta)^{\sum_{n=1}^N (1 - x_n)} \theta^{\sum_{n=1}^N z_n} (1 - \theta)^{\sum_{n=1}^N (1 - z_n)} p^{\sum_{n=1}^N f_n} (1 - p)^{\sum_{n=1}^N (1 - f_n)} \\
 &= \theta^{N_x} (1 - \theta)^{(N - N_x)} \theta^{N_z} (1 - \theta)^{(N - N_z)} p^{N_f} (1 - p)^{N - N_f} \\
 &= \theta^{N_x + N_z} (1 - \theta)^{(2N - (N_x + N_z))} p^{N_f} (1 - p)^{N - N_f} \\
 &= p(\mathbf{s}, \mathbf{z}, \mathbf{f} \mid \theta, p)
 \end{aligned}$$

Observe that  $N_x = N_x(\mathbf{s}, \mathbf{z}, \mathbf{f})$ ,  $N_z = N_z(\mathbf{z})$ , and therefore

$$p(\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s} \mid \theta, p) = p(\mathbf{s}, \mathbf{z}, \mathbf{f} \mid \theta, p).$$

Also,

$$p(\mathbf{s} \mid \theta, p) = \sum_{\mathbf{x}} \sum_{\mathbf{z}} \sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s} \mid \theta, p) = \sum_{\mathbf{f}} \sum_{\mathbf{z}} p(\mathbf{s}, \mathbf{z}, \mathbf{f} \mid \theta, p),$$

and

$$\begin{aligned}
 p(\theta \mid \mathbf{s}, p) &= \frac{p(\mathbf{s}, \theta, p)}{p(\mathbf{s}, p)} = \frac{p(\mathbf{s} \mid \theta, p) p(\theta, p)}{p(\mathbf{s}, p)} \\
 &= \frac{p(\mathbf{s} \mid \theta, p) p(\theta \mid p) p(p)}{p(\mathbf{s} \mid p) p(p)} \\
 &= \frac{p(\mathbf{s} \mid \theta, p) p(\theta \mid p)}{p(\mathbf{s} \mid p)} \\
 &= \frac{p(\mathbf{s} \mid \theta, p) p(\theta)}{p(\mathbf{s} \mid p)}.
 \end{aligned}$$

since  $\theta \perp\!\!\!\perp p$ .

## Box 5: Decision Rule

$$I\left(\frac{p(\theta_1 | \mathbf{s}, p)}{p(\theta_2 | \mathbf{s}, p)} > 1\right) \Rightarrow \text{choose } \theta_1,$$

which is equivalent to

$$I\left(\frac{p(\mathbf{s} | \theta_1, p)p(\theta_1)}{p(\mathbf{s} | \theta_2, p)p(\theta_2)} > 1\right) \Rightarrow \text{choose } \theta_1,$$

since

$$p(\theta | \mathbf{s}, p) \propto p(\mathbf{s} | \theta, p)p(\theta) = \sum_{\mathbf{f}} \sum_{\mathbf{z}} p(\mathbf{s}, \mathbf{z}, \mathbf{f} | \theta, p)p(\theta).$$

Notice that

$$p(\mathbf{s} | p) = \sum_{\theta} \sum_{\mathbf{f}} \sum_{\mathbf{z}} p(\mathbf{s}, \mathbf{z}, \mathbf{f} | \theta, p)p(\theta),$$

with  $\theta \in \{\theta_1, \theta_2\}$ .

These equations lead to the following results regarding question 3.

$$\frac{p([2, 2, 2, 2, 2, 0, 0, 0, 0] | \frac{2}{3}, \frac{1}{2})p(\frac{2}{3})}{p([2, 2, 2, 2, 2, 0, 0, 0, 0] | \frac{1}{2}, \frac{1}{2})p(\frac{1}{2})} = 1.80 \Rightarrow \theta_2 = \frac{2}{3}.$$

$$\frac{p([1, 0, 1, 0, 1, 0, 1, 0, 1, 0] | \frac{2}{3}, \frac{1}{2})p(\frac{2}{3})}{p([1, 0, 1, 0, 1, 0, 1, 0, 1, 0] | \frac{1}{2}, \frac{1}{2})p(\frac{1}{2})} = 0.13 \Rightarrow \theta_1 = \frac{1}{2}.$$

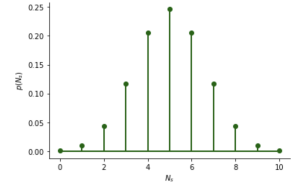


Figure 2: Probability distribution  $p(N_s)$  in Box for  $\theta = \frac{1}{2}$  for problem 3 in Box 9. The expected value  $\mu_{N_s} = 5$  and the entropy is  $E(N_s) = 2.71$ .

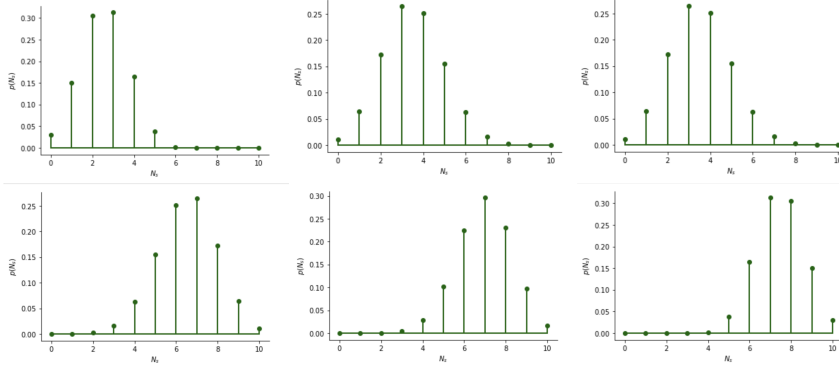


Figure 3: From left to right and from top to bottom, probability distributions  $p(N_s)$  in Box for  $\theta = \frac{1}{100}$ ,  $\theta$  are  $\frac{1}{9}$ ,  $\frac{1}{5}$ ,  $\frac{99}{100}$ ,  $\frac{8}{9}$ ,  $\frac{4}{5}$  for problem 3 in Box 9. The corresponding mean values,  $\mu_{N_s}$ , are 2.55, 3.06, 3.5, 6.5, 6.9, respectively. and  $\mu_{N_s} = 7.45$ , respectively. The corresponding entropies,  $E(N_s)$ , are 2.44, 2.56, 2.56, 2.44, and 2.22, respectively.

Also, notice that

$$E[\mathbf{s} | \theta, p] = \sum_{\mathbf{s}} \sum_{\mathbf{f}} \sum_{\mathbf{z}} \mathbf{s} p(\mathbf{s}, \mathbf{z}, \mathbf{f} | \theta, p),$$

$$E[N_s | \theta, p] = \sum_{N_s} N_s p(N_s | \theta, p).$$

The effect of  $\theta$  on  $N_s$  is

$$e(\theta_1, \theta_2 | p) = E[N_s | \theta_1, p] - E[N_s | \theta_2, p]$$

### Definition 2: Entropy

The average amount of information of a discrete random variable  $X$  is the expectation of  $I(x) = -\log_2(p(x))$  with respect to the distribution  $p(x)$  and is given by

$$H(X) = E[I(x)] = E[-\log_2(p(x))] = -\sum_x p(x) \log_2(p(x)).$$

The entropy of a Bernoulli random variable  $X \sim \text{Bernoulli}(x | p)$  is

$$H(X | p) = -p \log_2(p) - (1-p) \log_2(1-p).$$

### Kullback–Leibler Divergence

The Kullback–Leibler divergence (also called KL-divergence, relative entropy, and  $I$ -divergence) is a measure of how one probability distribution  $P_1$  is different from a second, reference probability distribution  $P_2$ . For discrete probability distributions,  $P_1$  and  $P_2$  defined on the same sample space, the relative entropy from  $P_2$  to  $P_1$  is defined to be

$$D_{KL}(P_1 \| P_2) = -\sum_x p_1(x) \log_2\left(\frac{p_2}{p_1}\right).$$

In other words, it is the expectation of the logarithmic difference between the probabilities  $P_1$  and  $P_2$ , where the expectation is taken using the probabilities  $P_1$ .

### Conjugate Priors

Let  $X$  and  $\Theta$  be two random variables. In Bayesian probability theory, if the posterior distribution  $p(\theta|x)$  is in the same probability distribution family as the prior probability distribution  $p(\theta)$ , the prior and posterior are then called conjugate distributions, and the prior is called a *conjugate prior* for the likelihood function  $p(x|\theta)$ . Recall that:

$$\underbrace{p(\theta | x)}_{\text{posterior}} = \frac{\overbrace{p(x | \theta)}^{\text{likelihood}} \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(x)}_{\text{evidence}}},$$

where

$$p(x) = \underbrace{\int p(x | \theta) p(\theta) d\theta}_{\text{marginalization}}.$$

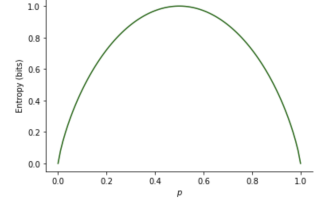


Figure 4: Entropy of the random variable  $X \sim \text{Bernoulli}(x | p)$ .

For example, if  $p(\theta)$  has a Beta distribution, and  $p(x | \theta)$  has a binomial distribution, then  $p(\theta | x)$  also has a Beta distribution. More specifically,

$$\begin{aligned} p(x | \theta) &\sim \text{Binomial}(x, |N, p) \\ p(\theta | \alpha, \beta) &\sim \text{Beta}(\theta | \alpha, \beta) \\ p(\theta | x) &\sim \text{Beta}(x | \alpha + \sum_{n=1}^N x_i, \beta + N - \sum_{n=1}^N x_i) \end{aligned}$$

Consider the probability distribution from Box 9:

$$p(\mathbf{s}, \mathbf{z}, \mathbf{f} | \theta, p) = \theta^{N_x + N_z} (1 - \theta)^{(2N - (N_x + N_z))} p^{N_f} (1 - p)^{N - N_f}$$

Clearly,

$$p(\theta, p | \mathbf{s}, \mathbf{z}, \mathbf{f}) = \text{Beta}(\theta | \alpha + \alpha_1, \beta + \beta_1) \text{Beta}(p | \gamma + \gamma_1, \delta + \delta_1)$$

where  $\alpha_1 = N_x + N_z$ ,  $\beta_1 = N - (N_x + N_z)$ ,  $\delta_1 = N_f$ ,  $\gamma_2 = N - N_f$ ,  $p(\theta | \alpha, \beta) = \text{Beta}(\theta | \alpha, \beta)$  and  $p(s | \gamma, \delta) = \text{Beta}(s | \gamma, \delta)$ .

Marginalizing with respect to  $p$ ,  $\mathbf{z}$ , and  $\mathbf{f}$  we obtain:

$$\begin{aligned} p(\theta | \mathbf{s}) &= \sum_{\mathbf{z}} \sum_{\mathbf{f}} \text{Beta}(\theta | \alpha + \alpha_1, \beta + \beta_1) \int_p \text{Beta}(p | \gamma + \gamma_1, \delta + \delta_1) dp, \\ &= \sum_{\mathbf{z}} \sum_{\mathbf{f}} \text{Beta}(\theta | \alpha + \alpha_1, \beta + \beta_1). \end{aligned}$$

### Expected Value and Covariance Matrix of Random Vectors

Let  $X_1, X_2, \dots, X_N$  be  $N$  discrete random variables with joint probability distribution  $p(x_1, x_2, \dots, x_N)$ . Let  $\mathbf{X} = [X_1, X_2, \dots, X_N]^T$

- The expected value of  $\mathbf{X}$  is given by

$$\mathbb{E}[\mathbf{X}] = \sum_{\mathbf{x}} \mathbf{x} p(\mathbf{x}) = [\mathbb{E}[X_1], \mathbb{E}[X_2], \dots, \mathbb{E}[X_N]]^T.$$

- The *covariance* of two discrete random variables  $X$  and  $Y$  is given by

$$\text{cov}(X, Y) = \mathbb{E}[(X - \mu_x)(Y - \mu_y)] = \mathbb{E}[XY] - \mu_x \mu_y,$$

where

$$\mathbb{E}[XY] = \sum_{\omega_x \in \Omega_x} \sum_{\omega_y \in \Omega_y} X(\omega_x) Y(\omega_y) P(X = X(\omega_x), Y = Y(\omega_y)).$$

Using the notation defined in the previous section we can write

$$\mathbb{E}[XY] = \sum_x \sum_y xy p(x, y).$$

- The covariance of vector  $\mathbf{X}$  is defined as

$$\text{cov}(\mathbf{X}) = \begin{bmatrix} \text{cov}(X_1) & \text{cov}(X_1, X_2) & \dots & \text{cov}(X_1, X_N) \\ \text{cov}(X_1, X_2) & \text{var}(X_2) & \dots & \text{cov}(X_2, X_N) \\ \dots & \dots & \dots & \dots \\ \text{cov}(X_1, X_N) & \text{cov}(X_2, X_N) & \dots & \text{var}(X_N) \end{bmatrix}$$

- The covariance of the random vectors  $\mathbf{X}$  and  $\mathbf{Y}$  is defined as

$$\text{cov}(\mathbf{X}, \mathbf{Y}) = \begin{bmatrix} \text{cov}(X_1, Y_1) & \text{cov}(X_1, Y_2) & \dots & \text{cov}(X_1, Y_N) \\ \text{cov}(X_2, Y_1) & \text{cov}(X_2, Y_2) & \dots & \text{cov}(X_2, Y_N) \\ \dots & \dots & \dots & \dots \\ \text{cov}(X_N, Y_1) & \text{cov}(X_N, Y_2) & \dots & \text{cov}(X_N, Y_N) \end{bmatrix}$$

### The Multivariate Gaussian Distribution

The Gaussian, also known as the normal distribution, is a widely used model for the distribution of continuous variables. In the case of a single variable  $x$ , the Gaussian distribution (probability density function) can be written in the form

$$\mathcal{N}(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x - \mu)^2\right\},$$

where  $\mu$  is the mean and  $\sigma^2$  is the variance. For a  $N$ -dimensional vector  $\mathbf{x}$ , the multivariate Gaussian distribution takes the form

$$\mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{D}{2}}} \frac{1}{\|\boldsymbol{\Sigma}\|^{\frac{1}{2}}} \exp\left\{-(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right\},$$

where  $\boldsymbol{\mu}$  is a  $N$ -dimensional mean vector,  $\boldsymbol{\Sigma}$  is a  $N \times N$  covariance matrix, and  $\|\boldsymbol{\Sigma}\|$  denotes the determinant of  $\boldsymbol{\Sigma}$ .

The Gaussian distribution arises in many different contexts and can be motivated from a variety of different perspectives. For example, the Gaussian distribution arises when we consider the sum of multiple random variables. The central limit theorem (due to Laplace) tells us that, subject to certain mild conditions, the sum of a set of random variables, which is of course itself a random variable, has a distribution that becomes increasingly Gaussian as the number of terms in the sum increases. We can illustrate this by considering  $N$  variables  $X_1, \dots, X_N$  each of which has a uniform distribution over the interval  $[0, 1]$ , and then considering the density of the mean  $(X_1 + \dots + X_N)/N$ . For large  $N$ , this density tends to be a Gaussian density (Figures 31 and 32).

### The Central Limit Theorem

In practice, the convergence to a Gaussian as  $N$  increases can be rapid. One consequence of this result is that the binomial distribu-

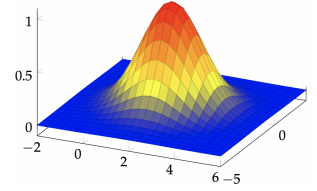


Figure 5: Surface plot of a two-dimensional Gaussian density.

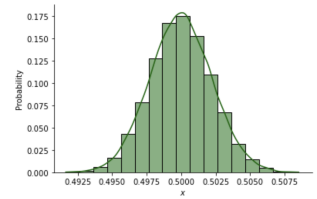


Figure 6: Histogram and KDE plots from 20,000 samples of  $(X_1 + \dots + X_N)/N$  for  $N = 50,000$ , with  $X_n \sim \text{Bernoulli}(x | p)$ . The estimated mean and variance are 0.49, and  $5.04e - 06$ , respectively.

tion, which is the sum of  $N$  observations of the Bernoulli random variable, will tend to a Gaussian as  $N \rightarrow \infty$  (Figure 31).

### Theorem 1: The Central Limit Theorem (CLT)

Let  $X_1, X_2, \dots, X_n$  be i.i.d. random variables with expected value  $E[X_i] = \mu < \infty$  and variance  $0 < \text{var}(X_i) = \sigma^2 < \infty$ . Then, the random variable

$$Z_N = \frac{\left[ \sum_{n=1}^N X_n \right] - \mu}{\sqrt{N}\sigma}$$

converges to the standard Gaussian (Normal) random variable as  $N \rightarrow \infty$ ,

$$\lim_{N \rightarrow \infty} F_{Z_N}(z) = \lim_{N \rightarrow \infty} P(Z_N \leq z) = \Phi(z), \quad \forall z \in \mathbb{R},$$

where  $\Phi(z)$  is the standard Gaussian cumulative distribution function

$$\Phi(z) = \int_{-\infty}^z \mathcal{N}(\alpha | 0, 1) d\alpha.$$

That is,

$$Z_\infty \sim \mathcal{N}(z | 0, 1).$$

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$$\bar{X}_N = \frac{1}{N} \sum_{n=1}^N X_n,$$

i.e. the average of  $X_1, X_2, \dots, X_N$ , has a Gaussian probability density with mean  $\mu$  and variance  $\frac{\sigma^2}{N}$ :

$$\bar{X}_N \sim \mathcal{N}(x | \mu, \frac{\sigma^2}{N}) \text{ as } N \rightarrow \infty.$$

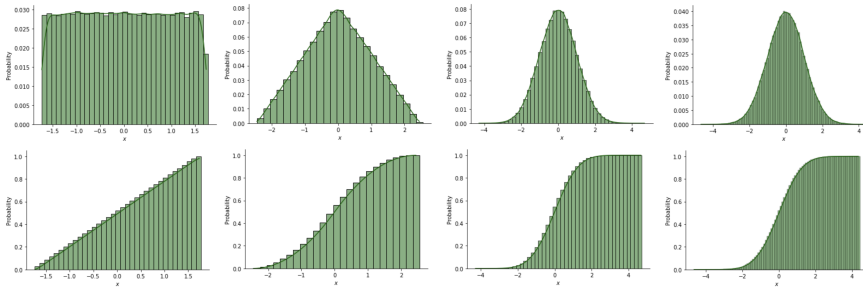


Figure 7: Top: Histogram and KDE plots of  $Z_N$  for various values of  $N$ : 1, 2, 10, and 100 (see Definition 15). We observe that as  $N$  increases, the density tends towards a Gaussian density. Bottom: Corresponding cumulative distribution functions. Here  $X_n \sim \text{Uniform}(x_n | 0, 1)$  for which  $\mu = 0.5$  and  $\sigma^2 = \frac{1}{12}$ .

Clearly,  $\lim_{N \rightarrow \infty} \bar{X}_N = \mu$ , so the average is an *unbiased estimator* of the mean.

### Chebyshev's Inequality

Let  $X$  be a random variable with a finite expected value  $\mu$  and finite non-zero variance  $\sigma^2$ . Then for any real number  $k > 0$ ,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Only the case  $k > 1$  is useful. When  $k \leq 1$ , right-hand side  $\frac{1}{k^2} \geq 1$  and the inequality is trivial as all probabilities are  $\leq 1$ .

#### Box 6: Example

Suppose that an unbiased coin is thrown 100 times. What is the bound that the number of heads will be greater than 70 or less than 30?

- Let  $K$  be the number of heads. Because  $K$  has a binomial distribution with  $\mu = 0.5$ :
- $E[K] = N\mu = 50$ .
- $\text{var}[K] = N\mu(1 - \mu) = 25$ .
- The standard deviation is  $\text{std}[K] = \sqrt{\text{var}[K]} = \sqrt{25} = 5$ .
- The values 70 and 30 are 20 units from the average, which is 4 standard deviations (i.e.,  $20/5$ ).

$$P(|K - E[K]| \geq 4\text{std}[K]) \leq \frac{1}{4^2} = 0.0625.$$

- Repeat the previous exercise for a)  $\mu = 0.6$  and b)  $\mu = 0.4$ .

Here is the full LaTeX code for the summary of Markov Random Fields (MRFs), including key facts and examples:

### Markov Random Fields Overview

A **Markov Random Field (MRF)** is a probabilistic model that represents the joint distribution of a set of random variables with dependencies encoded by an undirected graph. Each node in the graph corresponds to a random variable, and edges indicate direct probabilistic interactions. The defining property of an MRF is that each variable is conditionally independent of all others given its neighbors.

Let  $X = \{X_1, X_2, \dots, X_n\}$  be a set of random variables defined over a graph  $G = (V, E)$ , where  $V$  is the set of nodes and  $E$  the set of edges. The local Markov property is expressed as:

$$P(X_i | X_{V \setminus \{i\}}) = P(X_i | X_{\mathcal{N}(i)})$$

where  $\mathcal{N}(i)$  denotes the set of neighbors of node  $i$ .

MRFs are typically defined using *clique potentials*, which are functions over subsets of variables forming cliques in the graph. The joint distribution over all variables is given by the **Gibbs distribution**:

$$P(X = x) = \frac{1}{Z} \prod_{C \in \mathcal{C}} \psi_C(x_C)$$

Here,  $\mathcal{C}$  is the set of all cliques in the graph,  $\psi_C(x_C)$  is the potential function over clique  $C$ , and  $Z$  is the *partition function*:

$$Z = \sum_x \prod_{C \in \mathcal{C}} \psi_C(x_C)$$

## Examples of MRFs

### 1. Image Denoising

In image processing, each pixel is modeled as a node in a grid graph. The value of each pixel (e.g., grayscale intensity) is a random variable. Neighboring pixels are connected, and the MRF captures the spatial dependencies. The goal is to infer the true image from a noisy observation by maximizing the posterior distribution:

$$P(X | Y) \propto P(Y | X)P(X)$$

where  $Y$  is the observed noisy image and  $X$  is the latent clean image modeled as an MRF.

### 2. Ising Model

The Ising model is a classic example of an MRF used in statistical physics. Each node represents a spin variable  $X_i \in \{-1, +1\}$ , and edges represent interactions between neighboring spins. The joint distribution is:

$$P(X) = \frac{1}{Z} \exp \left( \sum_{(i,j) \in E} \beta X_i X_j + \sum_{i \in V} h_i X_i \right)$$

where  $\beta$  controls the interaction strength and  $h_i$  is an external field.

### 3. Semantic Segmentation

In computer vision, MRFs are used for semantic segmentation, where each pixel is assigned a label (e.g., sky, road, building). The MRF encodes the likelihood that neighboring pixels share similar labels and combines local evidence with spatial smoothness constraints.